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ESCUELA TÉCNICA
SUPERIOR DE INGENIEROS
INFORMÁTICOS

Image recreation using Genetic Algorithms and Color metrics

A metaheuristic optimization towards Mona Lisa

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"What if I told you that a computer could paint the Mona Lisa without ever being taught what art is? (and without using an LLM)"





Classic algorithms are like engineers—they plan everything, test everything, and need time.

Metaheuristics They're like adventurers. One tries random paths, another imitates the best, another dives down a risky valley hoping for a better view.

This study uses a GA to evolve randomly generated triangular shapes, optimizing their color (via CIE Lab $\Delta E * 94$)

The method balances exploration of geometric configurations with exploitation of high-fidelity solutions,

It demonstrates the potential of evolutionary computation to computationally approximate artistic masterpieces.



Introduction



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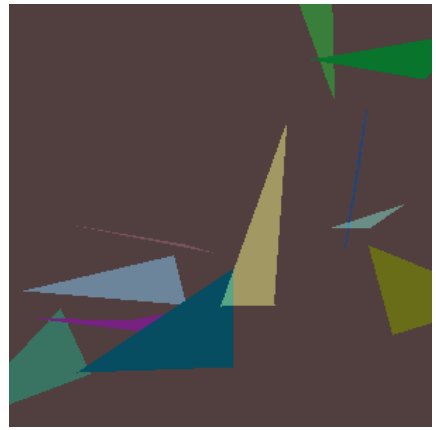
- **Objective**: Recreate the Mona Lisa using a genetic algorithm that optimizes color and structural similarity.
- **Motivation**: Explore the intersection of evolutionary computation and digital art.
- **Approach**: Use CIE Lab color metrics and genetic algorithms to generate images iteratively.

Methodology

Genetic Algorithm:

- **Initialization**: Randomly generate a population of images.
- **Selection**: Choose the fittest individuals based on color similarity.
- **Crossover**: Combine features of two parents to create offspring.
- **Mutation**: Introduce small random changes to the offspring.

Method



Example of a first individual.

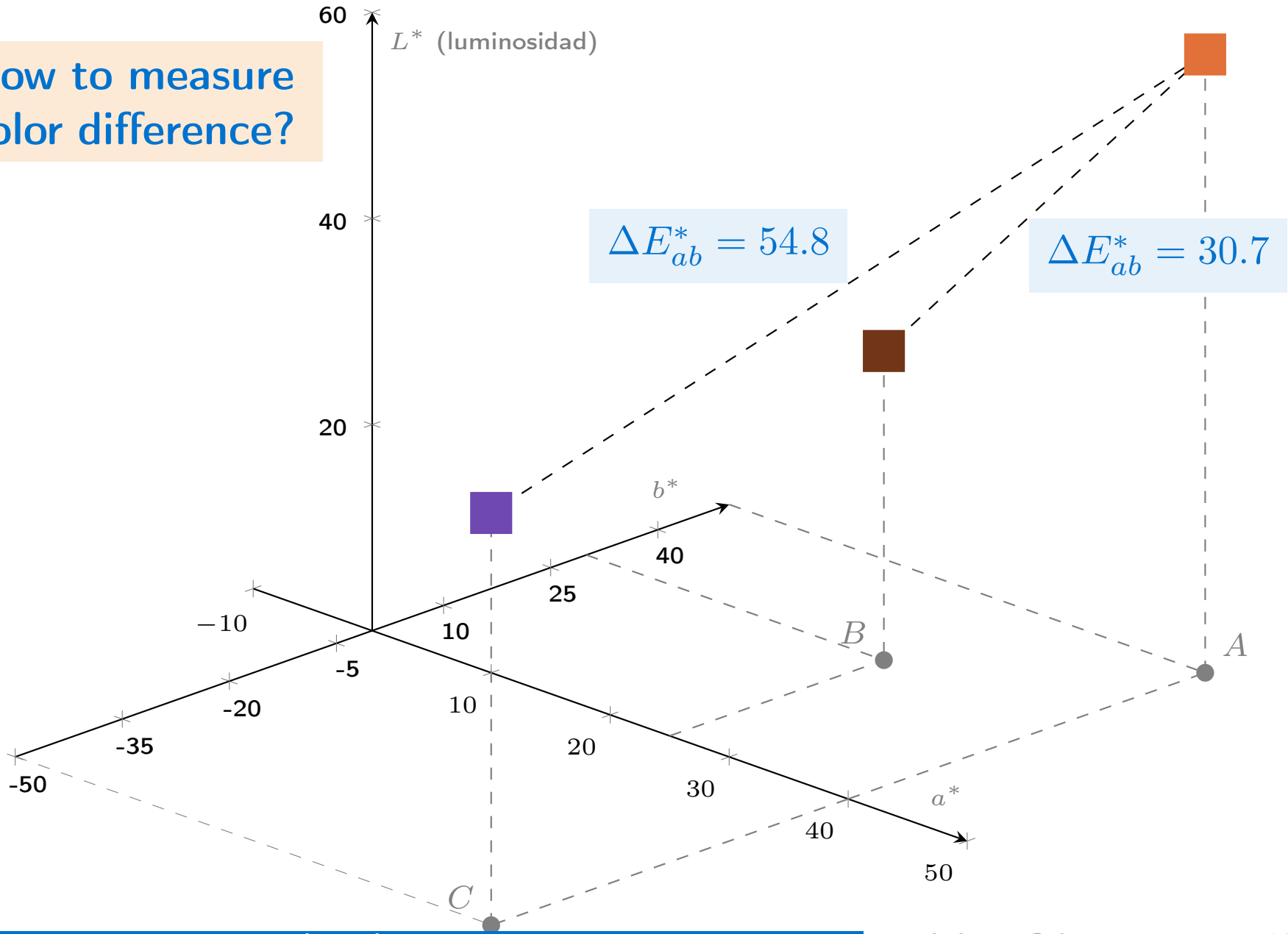
1. Each **individual** represents an image composed of triangles.
2. Each image is evaluated using the CIE Lab $\Delta CIE1994$[?] metric to measure **color similarity** as the mean value. Equation (??) is used in each pixel.
3. The GA **selects** (tournament), **crossover**, and **mutates** the best solutions to improve the approximation.
4. The process is repeated over several **generations** until an optimal solution is reached.
5. A stop condition is applied when fitness is < 0.01 for 3 consecutive individuals.

Color metrics

CIELAB-based color-difference ΔE

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How to measure color difference?



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Color metrics
Color coordinates

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- Use $\Delta E_{CIE1994}$ to measure color difference:

$$\Delta E^*_{94} = \sqrt{\left(\frac{\Delta L^*}{k_L S_L}\right)^2 + \left(\frac{\Delta C^*}{k_C S_C}\right)^2 + \left(\frac{\Delta H^*}{k_H S_H}\right)^2}$$

where:

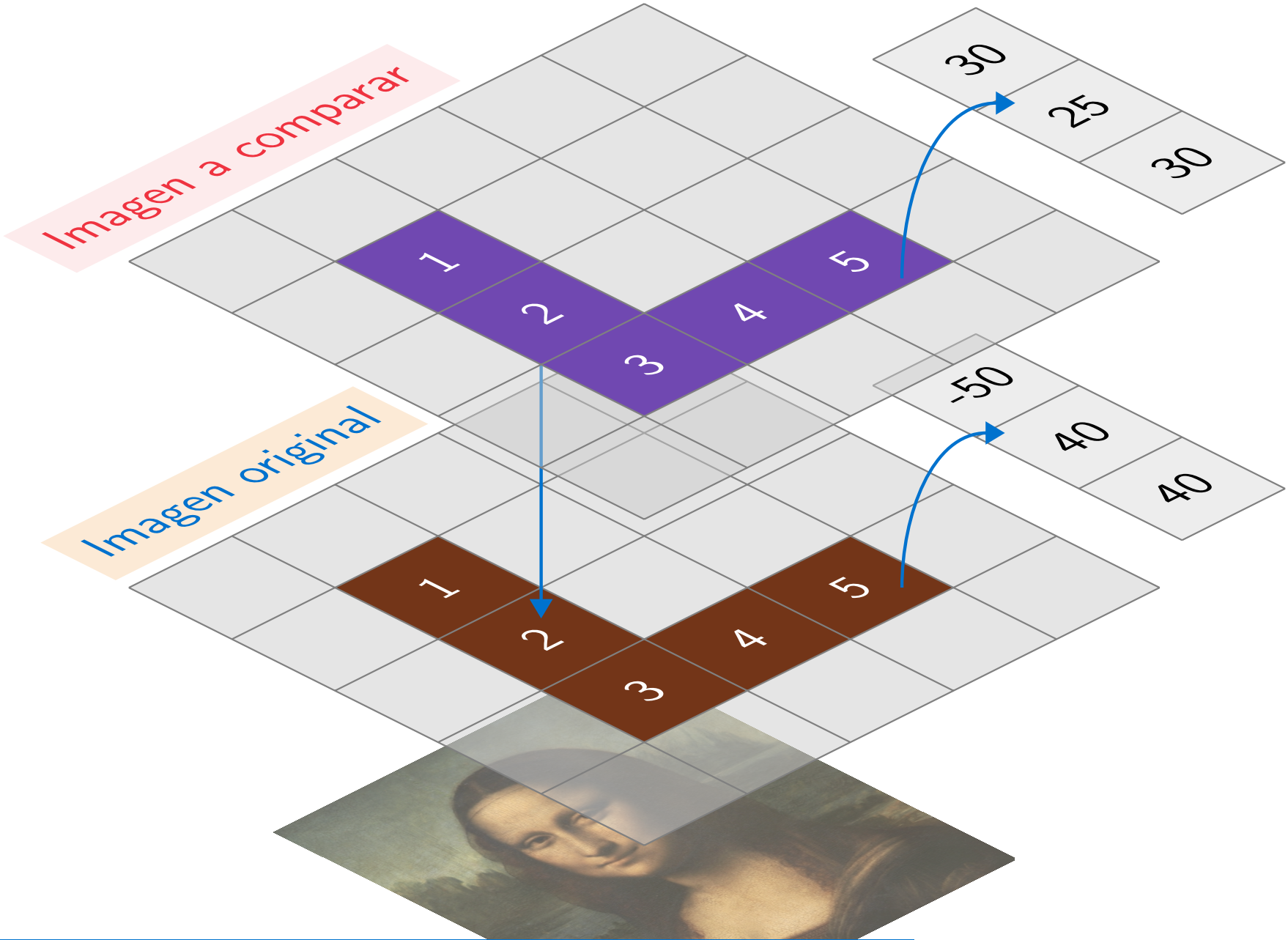
- $\Delta L^* = L^*_1 - L^*_2$ is luminosity difference.
- $\Delta C^* = C^*_1 - C^*_2$ is the color intensity difference.
- ΔH^* is the tone difference, calculated as:

$$\Delta H^* = \sqrt{(\Delta a^*)^2 + (\Delta b^*)^2 - (\Delta C^*)^2}$$

- k_L, k_C, k_H: Weighting factors.
- S_L, S_C, S_H: Scaling factors that depends on the observation conditions.

Color metrics

Copmaring images to evaluate fitness



Results

Generated Images

Evolution of the Mona Lisa over generations.

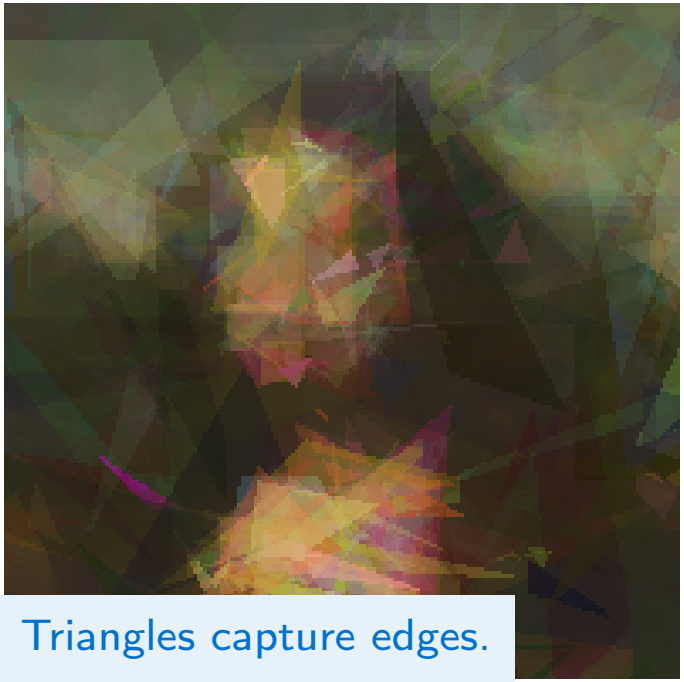
Comparison between the original and the final generated image.

Fitness Evolution

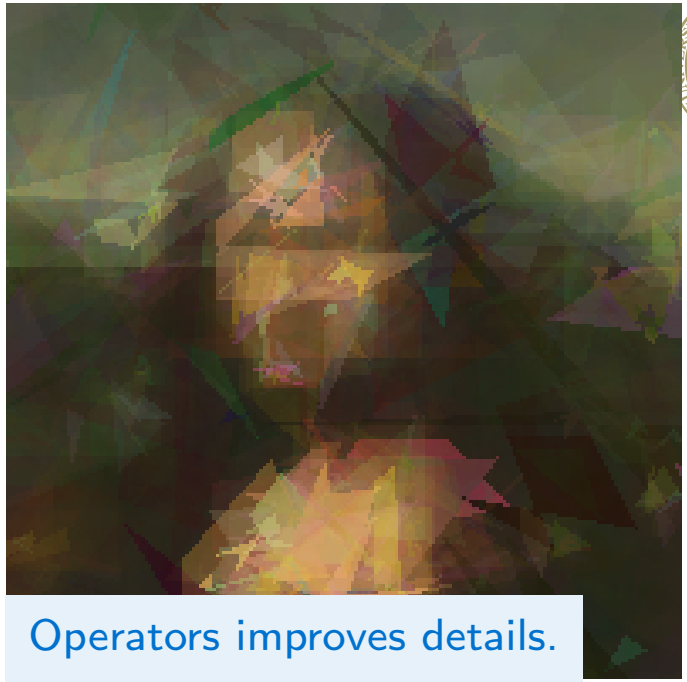
Graph showing the improvement of fitness over generations.

Discussion

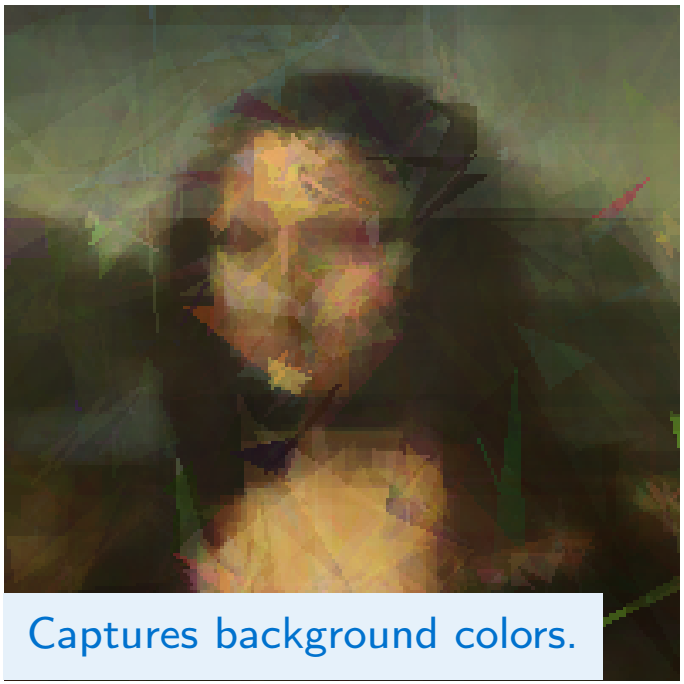
- **Challenges**:
  - High computational cost.
  - Sensitivity to outliers in color metrics.
- **Advantages**:
  - Effective convergence towards the target image.
  - Flexibility to use different color metrics.



Triangles capture edges.



Operators improves details.



Captures background colors.

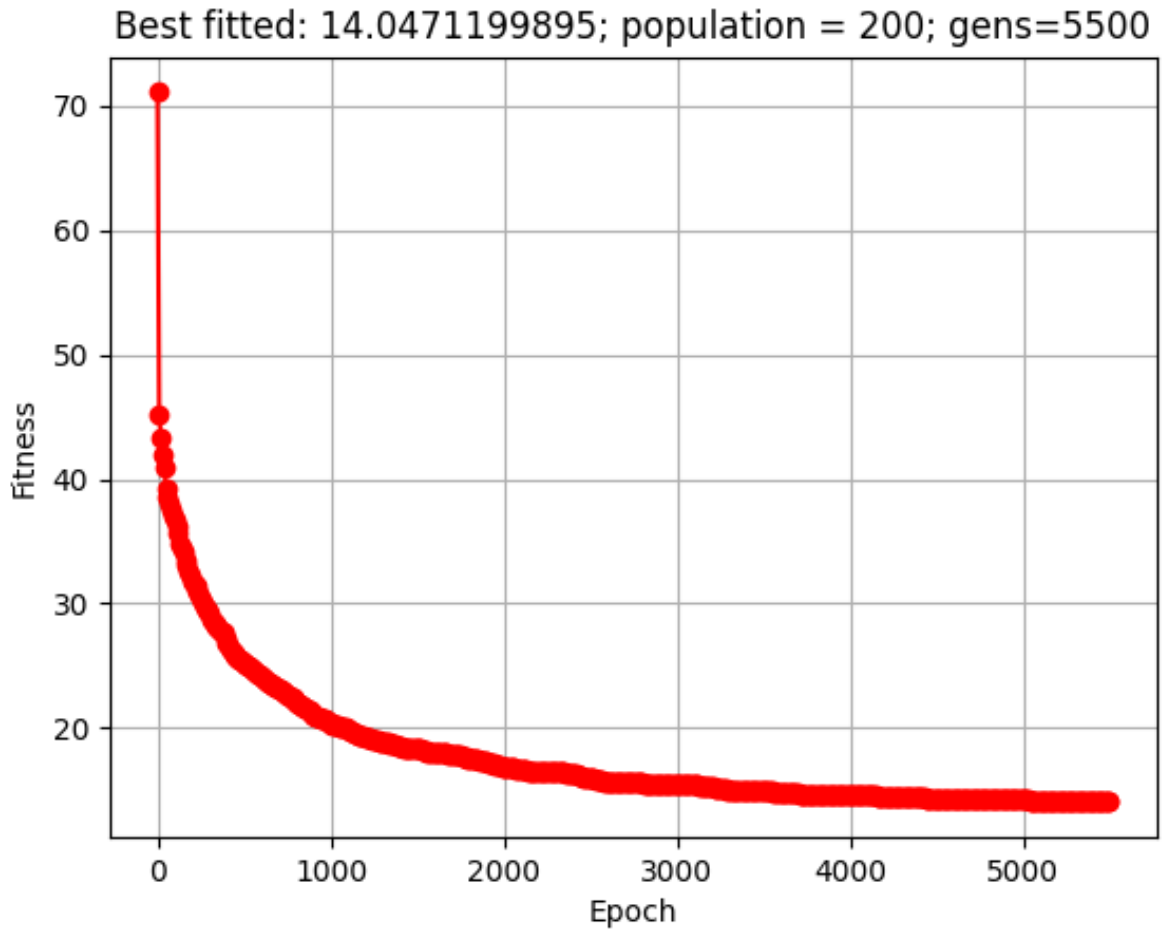


At 5,000 generations.

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Fitness evolution through generations

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Some caption.

Limitations

The algorithm showed rapid convergence in early generations (≤ 50), followed by stagnation, indicating a trade-off between exploration and exploitation.

Triangular shapes captured edges effectively but limited color blending. While $\Delta E\ CIE1994$ correlated well with human perception, it overlooked structural texture.

Larger populations (≥ 100) enhanced diversity at higher computational costs.

Conclusions

The genetic algorithm successfully approximated the Mona Lisa using triangular shapes and the $\Delta CIE1994$ color metric[?], achieving a 68 % reduction in color discrepancy over 100 generations.

While computationally intensive, the method demonstrated the potential of evolutionary strategies in artistic replication[?].

Future implementations could benefit from hybrid approaches (e.g., neural-guided mutations) and texture-aware metrics to overcome current limitations in smoothness and structural fidelity.